

1 A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.



1. Check the Charger

- Connect the charger properly.
- Check if the charger is working.
- **Checking:** Is the laptop getting power?

2. Check the Power Light

- Look for the charging or power light.
- **Checking:** Is the laptop receiving electricity?

3. Try Another Wall Socket

- Plug the charger into another working socket.
- **Checking:** Is the wall socket causing the problem?

4. Perform a Power Reset

- Unplug the charger.
- Hold the power button for **10–15 seconds**.
- Connect the charger again and try to turn on the laptop.
- **Checking:** Is a temporary power problem stopping the laptop from starting?

5. Check Battery and Hardware

- Try starting the laptop with the charger only, if possible.
- Check the battery and internal connections.
- **Checking:** Is the battery, RAM, motherboard, or another hardware component faulty?

2 You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.



Monitor Blank Screen – Easy Troubleshooting Steps

1. Check the monitor power

- Make sure the monitor is ON.
- Check if the power light is ON.
- Check the power cable and try another power socket.
- **Checking for:** A power or monitor problem.

2. Check the display cable

- Check the HDMI, VGA, or DisplayPort cable.
- Make sure the cable is connected properly on both sides.
- If possible, try another cable.
- **Checking for:** A loose or damaged cable.

3. Check the input source

- Open the monitor menu and select the correct input, such as HDMI or DisplayPort.
- **Checking for:** The monitor using the wrong input.

4. Check the PC connection

- Make sure the monitor cable is connected to the correct port on the computer.
- If there is a graphics card, connect the cable to the graphics card.
- **Checking for:** A wrong connection or graphics card problem.

5. Restart the computer

- Turn off the computer.
- Wait a few seconds and turn it ON again.
- Check if the computer logo or startup screen appears.

- **Checking for:** A temporary display or startup problem.

6. Check display settings

- If Windows is running, press **Windows + P**.
- Select **PC screen only** or **Duplicate**.
- Also check the brightness.
- **Checking for:** Wrong display settings or low brightness.

7. Try another monitor

- Connect the computer to another working monitor or TV.
- **Checking for:** Whether the problem is with the monitor or the computer.

8. Check internal hardware

- Turn off the computer before checking inside.
- Check if the RAM and graphics card are properly connected.
- **Checking for:** Loose or damaged internal parts.

Resolution

After these checks, we can find the faulty part. If another monitor works, the original monitor or cable may have a problem. If no monitor works, the computer's graphics card, RAM, or motherboard may have a problem.

3 A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.



Wireless Mouse Not Working

1. Check the Mouse

- Turn the mouse ON.
- Change the batteries if needed.
- Check if the mouse light is working.
- **Checking for:** Dead battery or damaged mouse.

2. Check the USB Receiver

- Remove the wireless receiver.
- Plug it into another USB port.

- Make sure it is connected properly.
- **Checking for:** Receiver or USB port problem.

3. Test the USB Port

- Plug another working USB device into the same port.
- **Checking for:** Whether the USB port is working.

4. Check Mouse Settings

- Open **Settings** → **Bluetooth & devices** → **Mouse**.
- Check if the mouse is detected.
- **Checking for:** Wrong mouse settings or driver problem.

5. Test the Mouse on Another Computer

- Connect the mouse receiver to another computer.
- Check if the mouse works.
- **Checking for:** If it works, the first computer may have a settings or driver problem. If it does not work, the mouse or receiver may be faulty.

Conclusion

- **Mouse does not work on any computer** → Mouse or receiver problem.
- **USB port does not work with other devices** → USB port problem.
- **Mouse works on another computer** → System settings or driver problem.

4 Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.



Gaming PC Overheating – 3 Hardware Causes

1. Fan Problem

- Check if the CPU and case fans are spinning.
- Listen for strange fan sounds.
- **Checking for:** A broken or slow fan.

2. Too Much Dust

- Turn OFF the PC and unplug it.
- Open the case and check the fans and heatsink for dust.
- Clean the dust carefully.
- **Checking for:** Dust blocking the airflow.

3. Old Thermal Paste

- Check the thermal paste between the CPU and heatsink.
- If it is dry, replace it with new thermal paste.
- **Checking for:** Poor cooling between the CPU and heatsink.

Conclusion

The PC may be overheating because of a **bad fan, dust, or old thermal paste.**

